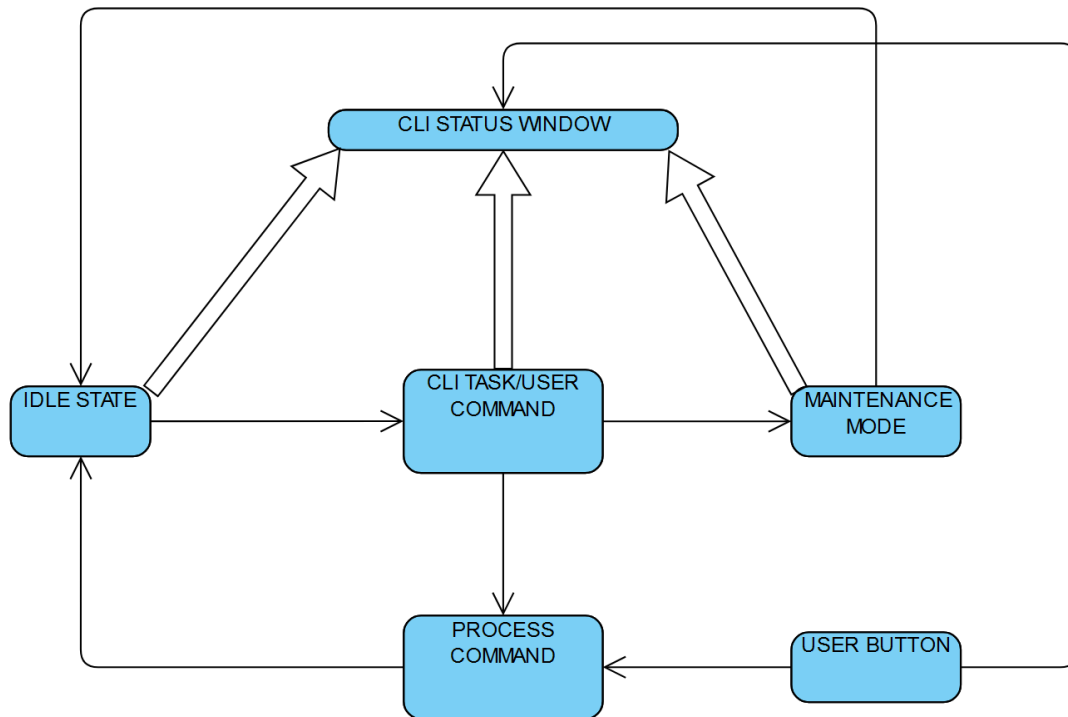


ENEL 452 TERM PROJECT – Elevator Control

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Task diagram: elevator in free rtos

The above Task diagram explains my elevator control system in Free Rtos. I will be implementing the tasks as follows:

CLI STATUS WINDOW: This window acts as my display screen for terminal. All operations are displayed here. I plan to implement different font colors for different floors. For e.g.: FLOOR 1, FLOOR 2, FLOOR 3 etc. Some other display examples include EMERGENCY STOP ACTIVATED, MAINTAINENCE MODE etc.

IDLE: This shows the elevator in idle state where it asks for user prompt in command line interface. This is the main state where the system comes back to after each process.

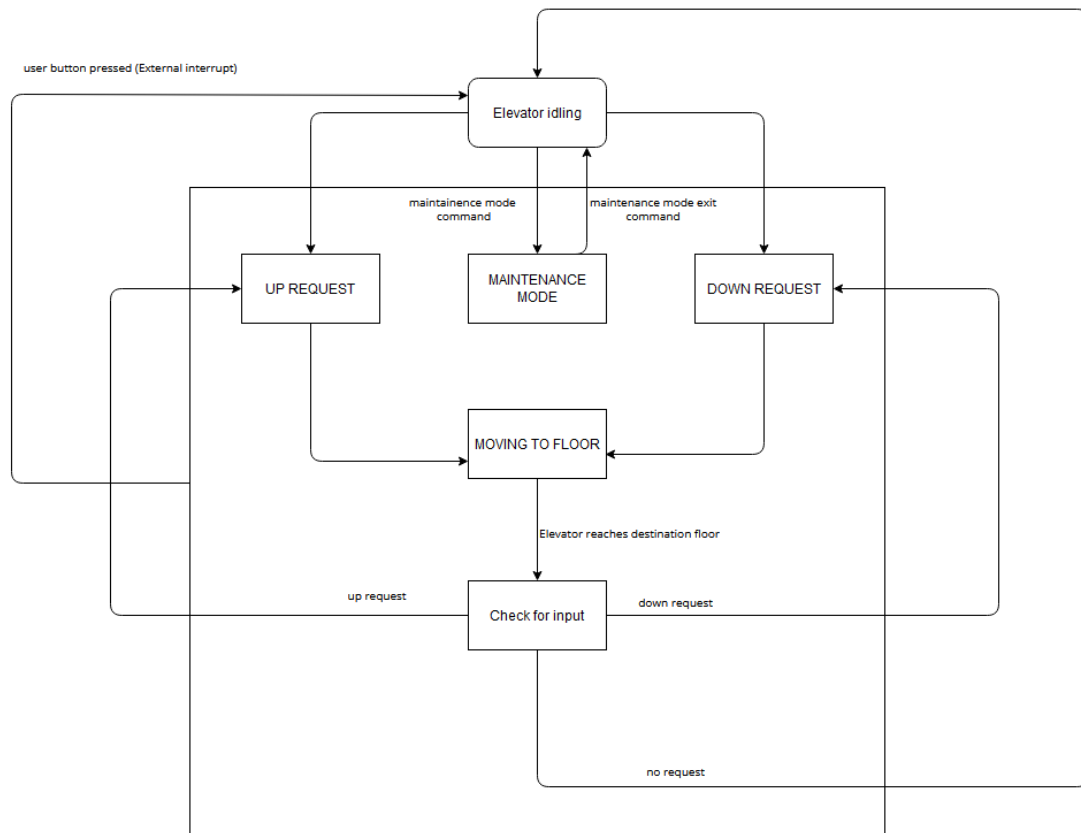
CLI TASK/USER COMMAND: This state implements the sending and receiving of functions using usart2 handler. The user can give command in this CLI interface to operate the elevator. Some commands include Go up, Go down, maintenance mode etc. All the user tasks will also be displayed in the CLI STATUS WINDOW which shows the program execution has started. Additionally, if user types maintenance as input, then the state changes to MAINTAINENCE MODE.

MAINTAINENCE MODE: This state is accessed by the user when they type maintenance mode command. When this command is run the elevator stops operating and the user will need to enter a different command to exit maintenance mode and go back to IDLE State.

PROCESS COMMAND: I plan to implement a timer in this block. The main function of this block is to take user input and operate the elevator and get back to the IDLE State. The timer delay would be our travel time for taking the elevator to each floor. (Let's say 5s per floor). Apart from processing commands this state also checks for interrupts and if an interrupt is identified the process stops right away and goes back to IDLE State right away. (USER BUTTON)

USER BUTTON: This task acts as my external interrupt and works as an emergency stop button. When the interrupt happens the process command should immediately revert back to idle state giving this ISR the highest priority. The user button once pressed also shows up on CLI STATUS WINDOW as “emergency stop activated”.

The state diagram for the Elevator system is as follows:



State diagram elevator system